



MATHEMATICS (EXTENSION 1)

2014 HSC Course Assessment Task 1

December 5, 2013

General instructions

- Working time – 50 minutes.
(plus 5 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets + answer sheet used in the correct order within this paper and hand to examination supervisors.

SECTION I

- Mark your answers on the answer sheet provided (numbered as page 5)

SECTION II

- Commence each new question on a new page. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

STUDENT NUMBER: **# BOOKLETS USED:**

Class (please ✓)

☐ 12M3A – Mr Zuber

☐ 12M4A – Ms Ziazaris

☐ 12M3B – Mr Berry

☐ 12M4B – Mr Lam

☐ 12M3C – Mr Lowe

☐ 12M4C – Mr Ireland

Marker's use only.

QUESTION	1-5	6	7	8	9	10	Total	%
MARKS	$\frac{5}{5}$	$\frac{8}{8}$	$\frac{6}{6}$	$\frac{7}{7}$	$\frac{5}{5}$	$\frac{6}{6}$	$\frac{37}{37}$	

Section I: Objective response

Mark your answers on the multiple choice sheet provided.

Marks

1. What is the equation of the parabola with focus $S(4, 0)$ and directrix $x = -2$? **1**

(A) $y^2 = 24(x - 4)$

(C) $y^2 = -12(x - 1)$

(B) $y^2 = -24(x - 4)$

(D) $y^2 = 12(x - 1)$

2. Which of the following would represent solutions to $\cos 2x + \sin x = 0$? **1**

(A) $\sin x = 0, \cos x = -\frac{1}{2}$

(C) $\sin x = \frac{1}{2}, \sin x = -1$

(B) $\sin x = -\frac{1}{2}, \sin x = 1$

(D) $\sin x = 0, \cos x = -1$

3. Which of the following calculations would return the acute angle between the lines $x - 2y = 6$ and $y = 3x - 1$? **1**

(A) $\tan \theta = \left| \frac{3 - 2}{1 + 6} \right|$

(C) $\tan \theta = \left| \frac{3 + 2}{1 - 6} \right|$

(B) $\tan \theta = \left| \frac{3 + \frac{1}{2}}{1 - \frac{3}{2}} \right|$

(D) $\tan \theta = \left| \frac{3 - \frac{1}{2}}{1 + \frac{3}{2}} \right|$

4. When the polynomial $P(x) = kx^3 + x^2 - (2k - 1)x + 2$ is divided by $(x + 1)$, the remainder is 4. **1**

What is the value of k ?

(A) -2

(B) 0

(C) 2

(D) 4

5. Which of the following represent all values of θ , for which $\tan \theta = \cot \theta$? **1**

(A) $\theta = n\pi \pm \frac{\pi}{4}$

(C) $\theta = n\pi + \frac{\pi}{4}$

(B) $\theta = 2n\pi \pm \frac{\pi}{4}$

(D) $\theta = n\pi - \frac{\pi}{4}$

End of Section I.
Examination continues overleaf.

Question 6 (8 Marks)		Commence a NEW page.	Marks
(a)	i. Show that $\frac{1 + \cos 2A}{\sin 2A} = \cot A$.		3
	ii. Hence find the exact value of $\cot 15^\circ$ in simplest form.		2
(b)	If $\cos \theta = 0.8$, $-\frac{\pi}{2} \leq \theta \leq 0$, find the exact value of $\cos \frac{\theta}{2}$.		3

Question 7 (8 Marks)		Commence a NEW page.	Marks
(a)	Derive the equation of the tangent to the parabola $x^2 = 4ay$ at the point $P(2ap, ap^2)$.		2
(b)	Given Q is the point $(2aq, aq^2)$, O is the origin, show that if OQ is parallel to the tangent, then $q = 2p$.		1
(c)	If M is the midpoint of PQ , find the equation of the locus of M as P and Q vary along the parabola such that OQ remains parallel to the tangent at P .		3

Question 8 (7 Marks)		Commence a NEW page.	Marks
(a)	The equation $x^3 - mx + 2 = 0$ has two equal roots.		
	i. Write down the expressions for the sum of the roots and for the product of the roots.		2
	ii. Hence find the value of m .		2
(b)	If α , β and γ are roots of the equation $x^3 - x^2 + 4x - 1 = 0$, find the value of $(\alpha + 1)(\beta + 1)(\gamma + 1)$.		3

Question 9 (8 Marks) Commence a NEW page. **Marks**

(a) From what external point are the tangents to the parabola $x^2 = 4y$ to be drawn such that the equation of the chord of contact is $2y - 3x + 2 = 0$? **2**

(b) Given $A(4, 2)$ and $B(-2, -8)$, show that the locus of the point $P(x, y)$ moving such that $\angle APB$ is a right angle, is given by **3**

$$x^2 - 2x + y^2 + 6y = 24$$

Question 10 (8 Marks) Commence a NEW page. **Marks**

Two of the roots of the equation $x^3 + ax^2 + b = 0$, ($a, b \in \mathbb{R}$ and non zero) are reciprocals of each other.

(a) Show that the third root is $-b$. **1**

(b) Show that $a = b - \frac{1}{b}$. **3**

(c) Show that the two roots, which are reciprocals, will be real if $-\frac{1}{2} \leq b \leq \frac{1}{2}$. **2**

End of paper.

Answer sheet for Section I

Mark answers to Section I by fully blackening the correct circle, e.g. “●”

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1 – ☐ A ☐ B ☐ C ☐ D

2 – ☐ A ☐ B ☐ C ☐ D

3 – ☐ A ☐ B ☐ C ☐ D

4 – ☐ A ☐ B ☐ C ☐ D

5 – ☐ A ☐ B ☐ C ☐ D

Suggested Solutions

Section I

1. (D) 2. (B) 3. (D) 4. (C) 5. (A)

Section II

Question 5 (Berry)

(a)

Question 6 (Berry)

(a)

Question 7 (Ziazari/Lam)

(a) (2 marks)

$$x^2 = 4ay \Rightarrow y = \frac{x^2}{4a}$$

$$\frac{dy}{dx} = \frac{2x}{4a} = \frac{x}{2a}$$

When $x = 2ap$,

$$\frac{dy}{dx} = \frac{2ap}{2a} = p$$

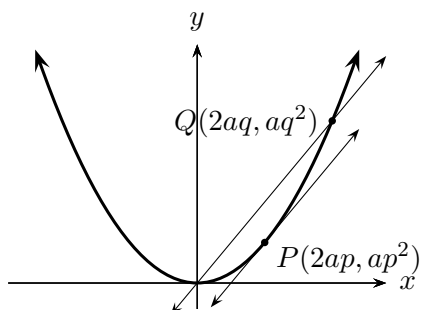
Applying point gradient formula,

$$y - ap^2 = p(x - 2ap)$$

$$y = px - 2ap^2 + ap^2$$

$$= px - ap^2$$

(b) (1 mark)



$$m_{OQ} = \frac{aq^2}{2aq} = \frac{q}{2}$$

Gradient of tangent at P is p . As $OQ \parallel$ tangent,

$$\frac{q}{2} = p$$

$$\therefore q = 2p$$

(c) (3 marks)

- ✓ [1] for midpoint.
- ✓ [1] for p in terms of x .
- ✓ [1] for final answer.

$$MP_{PQ} = \left(\frac{2ap + 2aq}{2}, \frac{ap^2 + aq^2}{2} \right)$$

$$= \left(a(p + q), \frac{a}{2}(p^2 + q^2) \right)$$

From the midpoint,

$$\begin{cases} x = a(p + q) \\ y = \frac{a}{2}(p^2 + q^2) \end{cases}$$

Since $q = 2p$,

$$\begin{cases} x = a(p + 2p) = 3ap \\ y = \frac{a}{2}(p^2 + 4p^2) = \frac{a}{2} \times 5p^2 \end{cases}$$

Rearrange the x equation and substitute into y :

$$p = \frac{x}{3a}$$

$$\therefore y = \frac{a}{2} \times 5 \left(\frac{x}{3a} \right)^2$$

$$= \frac{a}{2} \times 5 \times \frac{x^2}{9a^2}$$

$$= \frac{5x^2}{18a}$$

$$\therefore x^2 = \frac{18}{5}ay$$

Question 8 (Berry)

(a)

Question 9 (Berry)

(a)

①. D ②. B ③. D ④. C ⑤. A

2014 YR 12 ASSESSMENT TASK SOLUTIONS

EXT 1 TASK #1

$$\begin{aligned} \textcircled{6} a) \quad (i) \quad \text{LHS} &= \frac{1 + \cos 2A}{\sin 2A} \\ &= \frac{1 + 2\cos^2 A - 1}{2\sin A \cos A} \\ &= \frac{2\cos^2 A}{2\sin A \cos A} \\ &= \frac{\cos A}{\sin A} \\ &= \cot A \\ &= \text{RHS.} \end{aligned}$$

$$\begin{aligned} (ii) \quad \cot 15^\circ &= \frac{1 + \cos 30^\circ}{\sin 30^\circ} \\ &= \left(1 + \frac{\sqrt{3}}{2}\right) \div \frac{1}{2} \\ &= \frac{2 + \sqrt{3}}{2} \times \frac{2}{1} \\ &= 2 + \sqrt{3} \end{aligned}$$

$$\begin{aligned} b) \quad \cos \theta &= 0.8 \\ \cos 2\theta &= 2\cos^2 \theta - 1 \\ \therefore \frac{\cos 2\theta + 1}{2} &= \cos^2 \theta \\ \therefore \frac{\cos \theta + 1}{2} &= \cos^2 \frac{\theta}{2} \\ \frac{0.8 + 1}{2} &= \cos^2 \frac{\theta}{2} \\ 0.9 &= \cos^2 \frac{\theta}{2} \\ \cos \frac{\theta}{2} &= \pm \sqrt{0.9} \\ \text{But in 4th quad} \\ \therefore \cos \frac{\theta}{2} &= \sqrt{0.9} = \frac{3}{\sqrt{10}} \end{aligned}$$

$$7. a) \quad y = \frac{x^2}{4a}$$

$$\frac{dy}{dx} = \frac{x}{2a}$$

$$\text{At } x = 2ap$$

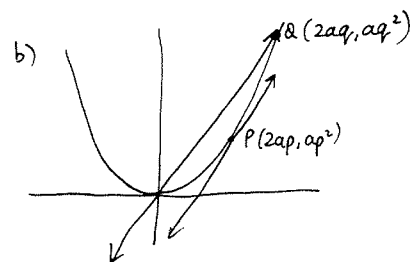
$$\frac{dy}{dx} = p$$

Eq'n of tangent:

$$y - ap^2 = p(x - 2ap)$$

$$y - ap^2 = px - 2ap^2$$

$$y = px - ap^2$$



$$\text{Slope } m_{OQ} = \frac{aq^2}{2aq} = \frac{q}{2}$$

$$m \text{ of tang at } P = p.$$

Since $OQ \parallel$ tangent at P then

$$p = \frac{q}{2}$$

$$\therefore 2p = q$$

$$\boxed{\text{OR.}} \text{ Eq'n of } OQ: y - aq^2 = p(x - 2aq)$$

$$y - aq^2 = px - 2apq$$

But $(0,0)$ satisfies eq'n

$$0 - aq^2 = 0 - 2apq$$

$$-q = -2p$$

$$\therefore 2p = q$$

$$\begin{aligned} c) \quad M &= \left(\frac{2ap + 2aq}{2}, \frac{ap^2 + aq^2}{2} \right) \\ &= \left(a(p+q), a \frac{(p^2 + q^2)}{2} \right) \end{aligned}$$

$$\text{i.e. } x = a(p+q)$$

$$\text{But } q = 2p$$

$$\therefore x = a3p$$

$$\therefore p = \frac{x}{3a}$$

$$\text{Sub p into } y = a \frac{(p^2 + q^2)}{2}$$

$$y = a \left(\frac{x^2}{9a^2} + (2p)^2 \right)$$

$$y = a \left(\frac{x^2}{9a^2} + 4 \left(\frac{x^2}{9a^2} \right) \right)$$

$$y = \frac{a}{2} \left(\frac{5x^2}{9a^2} \right)$$

$$\boxed{18ay = 5x^2}$$

8. a) $x^3 - mx + 2 = 0$

(i) Let the roots be α, α, β .

$$\therefore 2\alpha + \beta = 0 \quad \text{--- (1)}$$

$$\alpha^2\beta = -2 \quad \text{--- (2)}$$

(ii) $\beta = -2\alpha \quad \text{--- (3)}$

Sub (3) into (2)

$$\alpha^2(-2\alpha) = -2$$

$$\alpha^3 = 1$$

$$\therefore \alpha = 1$$

$$\therefore \beta = -2$$

$\therefore$ Roots 1, 1, -2

$$1 + -2 - 2 = -m$$

$$-3 = -m$$

$$\therefore m = 3$$

b) $x^3 - x^2 + 4x - 1 = 0$

$$\begin{aligned} (a+1)(b+1)(c+1) &= (ab+a+b+1)(c+1) \\ &= abc + ab + ac + a + bc + b + c + 1 \\ &= abc + (ab+ac+bc) + (a+b+c) + 1 \\ &= 1 + 4 + 1 + 1 \\ &= 7 \end{aligned}$$

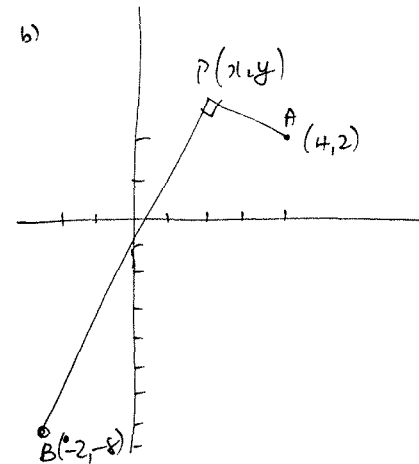
9. a) Chord of contact

$$xx_0 = 2a(y+y_0)$$

$$\begin{aligned} \text{ie. } xx_0 - 2xy - 2xy_0 &= 0 \\ + 3x - 2y - 2 &= 0 \end{aligned}$$

$$\therefore x_0 = 3, y_0 = 1$$

External point is (3,1)



Let P be the point (x,y)

$$m_{AP} = \frac{y-2}{x-4}$$

$$m_{PB} = \frac{y+8}{x+2}$$

Since $AP \perp PB$ then

$$\frac{y-2}{x-4} \times \frac{y+8}{x+2} = -1$$

$$y^2 + 8y - 2y - 16 = -(x^2 + 2x - 4x - 8)$$

$$y^2 + 8y - 2y - 16 = -x^2 - 2x + 4x + 8$$

(10) a) Equ'n of PQ : $m = m(0, -2)$

$$\begin{aligned} y+2 &= m(x-0) \\ y+2 &= mx \\ y &= mx-2. \end{aligned}$$

b) Sub $2t, t^2$
 $t^2 = m(2t) - 2$
 $t^2 - 2mt + 2 = 0.$

c) $p+q = 2m$
 $pq = 2$

$$R(-pq(p+q), (p+q)^2 - pq + 2)$$

$$\text{i.e. } R(-2(2m), (2m)^2 - 2 + 2)$$

$$R(-4m, 4m^2)$$

$x^2 = 4y$ is original parabola

$$\begin{aligned} \text{LHS} &= x^2 \\ &= (4m)^2 \\ &= 16m^2 \end{aligned}$$

$$\begin{aligned} \text{RHS} &= 4y \\ &= 4(4m^2) \\ &= 16m^2 \end{aligned}$$

(11) a) $x^3 + ax^2 + b = 0$

Let the roots be $\alpha, \frac{1}{\alpha}, \beta$

Product of roots : $\alpha \cdot \frac{1}{\alpha} \cdot \beta = -b$

$$\therefore \boxed{\beta = -b}$$

b) Sum of roots : $\alpha + \frac{1}{\alpha} + \beta = -a$
 $\alpha + \frac{1}{\alpha} - b = -a$

$$\therefore \alpha + \frac{1}{\alpha} = b - a$$

Sum of roots 2 at a time : $\alpha \cdot \frac{1}{\alpha} + \frac{1}{\alpha} \cdot \beta + \beta \alpha = 0$

$$1 + \frac{\beta}{\alpha} + \alpha\beta = 0$$

$$1 - \frac{b}{\alpha} - b\alpha = 0 \quad \text{As } \beta = -b$$

$$\begin{aligned} 1 - b\left(\frac{1}{\alpha} + \alpha\right) &= 0 \\ \frac{1}{\alpha} + \alpha &= \frac{1}{b} \end{aligned}$$

But $\alpha + \frac{1}{\alpha} = b - a$

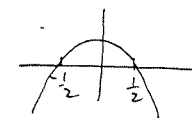
$$\therefore \frac{1}{b} = b - a$$

$$\therefore \boxed{a = b - \frac{1}{b}}$$

c) $\alpha, \frac{1}{\alpha}$ are roots of eq'n

$$x^2 - \left(\alpha + \frac{1}{\alpha}\right)x + \alpha \cdot \frac{1}{\alpha} = 0$$

$$x^2 - \frac{1}{b}x + 1 = 0 \quad (1)$$



Real when $\Delta \geq 0$.

$$\frac{1}{b^2} - 4 \geq 0$$

$$1 - 4b^2 \geq 0$$

$$(1-2b)(1+2b) \geq 0$$

$$\text{i.e. } \boxed{-\frac{1}{2} \leq b \leq \frac{1}{2}}$$

OR
 Sub in $P(-b) = 0$
 $-b^3 + ab^2 + b = 0$
 $ab^2 = b^3 - b$
 $a = b - \frac{1}{b}$

HIGH QUALITY MATHEMATICAL COMMUNICATION IS REQUIRED!

"Show" questions require clearly communicated mathematical reasoning. They are the mathematics analogue of a well crafted English essay. You are telling a story – it must have a clear beginning, clearly marked "paragraphs", with an argument presented in a logical sequence.

Reason
$$\begin{array}{l} A + C = B + C \\ A = B \end{array} \quad (1)$$

Reason
$$\begin{array}{l} 2A + Z = Q + Z \\ 2A = Q \end{array} \quad (2)$$

(1) + (2)
$$\begin{array}{l} 3A = B + Q \\ A = (B+Q)/3 \end{array}$$

In particular:

1. Each new equation introduced into the flow must have an explanation where it comes from. eg: "Sum of roots two at a time", "Substitute (1) into (2)"
2. Add labels (1), (2) or (A), (B) to the end of each block of reasoning so that you can refer to it when you start a new block of reasoning linking equations together.
3. Work in ONE column, going down the page.
4. Each line must have an equals sign – you are making a statement of truth and carrying this truth throughout the argument.

Running out of time, or 'working out' while exploring a solution is not an excuse to make a mess. If you expect it to be marked, then you owe it to the marker to clearly explain what you are doing.

It takes only a few extra seconds to label equations, to write a two word justification for a new equation. You are thinking it in your head anyway – so write it down.

MARKS CAN AND SHOULD BE DEDUCTED FOR POOR SETTING OUT.

STUDENT SAMPLE OF QUALITY WORK

Question 11

a) let roots be $\alpha, \frac{1}{\alpha}, \beta$

then product of roots are

$$\alpha \cdot \frac{1}{\alpha} \cdot \beta = -\frac{b}{c}$$

$$\beta = -b$$

$\therefore$ other root is $-b$

b) sum of roots 2 at a time

$$\alpha \cdot \frac{1}{\alpha} + \alpha\beta + \frac{\beta}{\alpha} = 0$$

$$1 + \alpha\beta + \frac{\beta}{\alpha} = 0$$

$$\alpha + \alpha^2\beta + \beta = 0 \quad (1)$$

sum of roots

$$\alpha + \frac{1}{\alpha} + \beta = -a$$

$$\alpha + \frac{1}{\alpha} - b = -a$$

$$\alpha + \frac{1}{\alpha} = -a + b \quad (2)$$

from (1): $1 + \alpha\beta + \frac{\beta}{\alpha} = 0$

$$1 + \beta(\alpha + \frac{1}{\alpha}) = 0$$

$$(2) \rightarrow (1)$$

$$1 + \beta(-a + b) = 0$$

$$1 - b(-a + b) = 0$$

$$-b(-a + b) = -1$$

$$-a + b = \frac{1}{b}$$

$$\therefore a = b - \frac{1}{b}$$

continued...